

Studying the User Adaptability to Hyperbolic Spaces and Delay Time Scenarios

Ana R. Rebelo*

Rui Nóbrega†

Fernando Birra‡

NOVA-LINCS, DI, NOVA School of Science and Technology
Universidade Nova de Lisboa, Portugal

Figure 1: The Tangible VR Lab is a virtual environment created to study user interaction with tangible objects under different conditions of time and space. As a case study, the first morphing scenarios relied on simple tasks: Target Game and Puzzle Game.

ABSTRACT

To create immersive virtual experiences, it is crucial to understand how users perceive Virtual Environments (VEs) and which interaction techniques are most appropriate for their tasks. We created a tangible VE — the VR Lab — where it is possible to study space and time conditions to analyse the user's adaptability to different forms of interaction. As a case study, we restricted the scope of the investigation to two morphing scenarios. The space morphing scenario compares the adaptability of users to Euclidean versus hyperbolic spaces. The time morphing scenario aims to establish from which values the visual delay affects performance.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality

1 INTRODUCTION

Human perception can vary in Virtual Reality (VR). The virtual space is usually underestimated [4], and time seems to pass faster. Understanding human perception and the system requirements allow important insights to design perceptually more accurate VEs. For instance, some interaction techniques take advantage of the human inability to detect minor spatial or temporal discrepancies. Redirected techniques [3, 8] introduce slight rotations to the user's trajectory to avoid real-world obstacles. The human inability to detect small time delays also allows relaxing low latency requirements [1].

*e-mail: ar.rebelo@campus.fct.unl.pt

†e-mail: rui.nobrega@fct.unl.pt

‡e-mail: fpb@fct.unl.pt

The VR Lab aims to be a study site to explore diverse spatiotemporal conditions. In this way, it is possible to establish a task and analyse it under different conditions to evaluate which one is more suitable according to pre-established parameters (e.g., performance, presence, embodiment). After preliminary research on the current challenges in VR, we found some scenarios that deserve to be further studied, including locomotion techniques [6], navigation in non-Euclidean spaces [5, 7] and tolerance to visual feedback delay [2]. As proof of concept, we chose two scenarios: a spatial morphing scenario and a time morphing scenario.

2 USER STUDY

As shown in Fig. 1, the VR Lab was designed to enable passive haptic feedback for more natural and intuitive interaction. The case study involved three tangible objects — a cube and two triangular prisms — 3D printed and marked on the upper face with target images. The Vuforia SDK tracks these planar images through the webcam to determine the position and orientation of each object. The objects had purposely simple shapes to avoid tracking errors. The tasks proposed at each scenario also had basic object manipulation (translation and rotation) to prevent bias from task complexity.

Spatial Morphing Scenario Although humans are used to Euclidean spaces, the non-Euclidean (e.g., hyperbolic) properties have revealed their applicability in games, large dataset visualisation and navigation in VR [5, 7] due to its constant negative curvature that creates the illusion of a wider space. This scenario further explored the applicability of the hyperbolic geometry in VR, analysing whether the user adaptation is equally easy as in Euclidian spaces. The virtual play area (i.e., table play area) was augmented by about three times compared to the physical table. To increase the play area, three approaches were studied: a linear (F1) and two hyperbolic (F2 and F3). Fig. 2 shows the conditions assessed. As a baseline, F0 has no distortion; the real and virtual play areas had the same dimension,

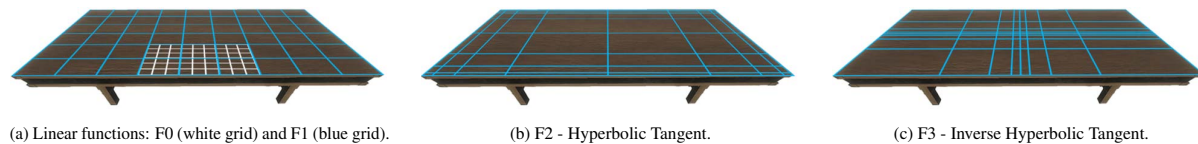


Figure 2: Simplified view of the functions studying in the space morphing scenario.

and objects moved in both spaces as in the normal real-world conditions. F1 preserves the Euclidean geometry, additionally applying a scaling factor so that it is possible to move objects along the entire virtual table. The hyperbolic function F2 makes the space wider in the center and tighter at the ends, translating movements of the physical objects into larger virtual movements when performed in the center of the play area. Under these conditions, the user must be more precise to place objects in the center since a minimal movement quickly brings the object to the edges. F3 cause the inverse behaviour. As a task, the user played the Target Game (Fig. 1), which consisted in getting each object into the target of the same colour.

Time Morphing Scenario This scenario is in line with the systematic study on human tolerance to visual feedback delay [2]. Visual delay was applied to the movement of the virtual tangible objects. The delay levels assessed were 0 ms, 20 ms, 50 ms, 100 ms, 200 ms, 500 ms and 1000 ms. As a task, the user had to play the Puzzle Game (Fig. 1), which consisted of using the tangible objects to assemble the displayed figures.

Subjects In this preliminary study, the participant pool consisted of 28 voluntary participants (20 male, 8 female) aged between 18 and 35 ($M=23.00$, $SD=3.66$). 16 participants reported never having experienced VR before, and 15 said they play video games regularly (daily or weekly).

Procedure All participants performed both scenarios in all conditions, following a Latin-square order between participants to avoid biased results by learning or fatigue factors. Performance (i.e., task completion time) and subjective ease were measured in each condition. Subjective ease was measured by asking the participant to rate, on a 7-point Likert scale (1=strongly disagree and 7=strongly agree), the statement "I found this task easy."

3 RESULTS

The data were analysed in both scenarios using the Friedman test for repeated measures and the Wilcoxon signed-rank tests for post hoc pairwise comparisons. The preliminary results are presented below.

Spatial Morphing Scenario There was a significant difference in performance ($\chi^2=64.76$, $p<.001$) and subjective ease ($\chi^2=57.07$, $p<.001$) between F0 and all morphing conditions. Comparing the linear function F1 with the hyperbolic functions, F1 revealed the better performance and more subjective ease than F2; F1 and F3 revealed no significant differences. In general, F2 showed to be the most challenging; however, all participants were able to adapt quickly to the hyperbolic spaces. The similarity between F1 and F3 was mentioned by users, who in both cases felt that they had more space to move objects around, even if the physical space remained the same the entire time. The F3 geometry allows reaching the ends of the table faster without losing performance. These preliminary results suggest the applicability of hyperbolic geometry to create spaces with different motion amplitude (e.g., reaching more quickly objects that are further away from the user), which may be difficult to recreate with Euclidean geometry alone.

Time Morphing Scenario The performance decreased significantly starting at the 200 ms delay ($Z=-3.48$, $p<.001$). However, subjective ease only began to decline significantly from the 500 ms delay ($Z=-3.90$, $p<.001$). Haptic feedback showed to help users at higher delays, as they were not solely dependent on the visual response. This insight can be helpful to relax the low latency requirements in situations where users have tactile information to complement the visual information. In the future, we are interested in exploring further the relationship between delay tolerance and users' experience with video games.

4 DISCUSSION AND CONCLUSION

Exploring space and time in VR helps understand human perception and find new ways of interaction. As the case study, non-Euclidean spaces can be effective in navigation. The hyperbolic geometry rejects the parallel postulate, making it possible to access more area within a given distance; it becomes possible to navigate naturally through complex spaces of infinite branching structures and dodge obstacles easily due to the hyperbolic constant negative curvature. Studying users' tolerance to delays allows establishing acceptable latency that an interactive application may have without compromising the viability of the task. We believe that the VR Lab concept can further explore many other current challenges in the HCI field.

ACKNOWLEDGMENTS

This work was supported by NOVA.ID.FCT through grant UIDB/04516/2020/TRA/BIL/26.

REFERENCES

- [1] R. Albert, A. Patney, D. Luebke, and J. Kim. Latency requirements for foveated rendering in virtual reality. *ACM Trans. Appl. Percept.*, 14, 2017. doi: 10.1145/3127589
- [2] K. Brunnström, E. Dima, T. Qureshi, M. Johanson, M. Andersson, and M. Sjöström. Latency impact on quality of experience in a virtual reality simulator for remote control of machines. *Signal Processing: Image Communication*, 89, 2020. doi: 10.1016/j.image.2020.116005
- [3] C. Carvalho, R. Nóbrega, H. da Silva, and R. Rodrigues. User redirection and direct haptics in virtual environments. In *Proc. MM*, p. 1146–1155. ACM, NY, USA, 2016. doi: 10.1145/2964284.2964293
- [4] M. Gonzalez-Franco, P. Abtahi, and A. Steed. Individual differences in embodied distance estimation in virtual reality. In *Proc. IEEE VR*, pp. 941–943, 2019. doi: 10.1109/VR.2019.8798348
- [5] V. Hart, A. Hawksley, E. Matsumoto, and H. Segerman. Non-euclidean virtual reality i: Explorations of h^3 . In *Proc. Bridges 2017: Mathematics, Art, Music, Architecture, Education, Culture*, pp. 33–40. Tessellations Publishing, Phoenix, Arizona, 2017.
- [6] E. Langbehn, P. Lubos, and F. Steinicke. Evaluation of locomotion techniques for room-scale vr: Joystick, teleportation, and redirected walking. In *Proc. VRIC*. ACM, 2018. doi: 10.1145/3234253.3234291
- [7] V. A. Pisani, O. Hurd, N. Hawthorne, and S. Kurniawan. Navigation by walking in hyperbolic space using virtual reality. In *Proc. CHI PLAY*, p. 611–618. ACM, 2019. doi: 10.1145/3341215.3356287
- [8] F. Steinicke, G. Bruder, J. Jerald, H. Frenz, and M. Lappe. Estimation of detection thresholds for redirected walking techniques. *IEEE Transactions on Visualization and Computer Graphics*, 16, 2010. doi: 10.1109/TVCG.2009.62